Design report

Solution Configurator, residential system

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Site and inputs

Postcode	64331 Weiterstadt
Position	49.904, 8.589
Annual consumption	5,000 kWh
PV array	10 kWp
Roof plane 1	10 kWp, tilt 35°, facing 180°
Grid connection	3-phase
Backup supply	Not needed
Heat pump	No
Electric car	No
Revision	Design data c6f964f0b599 · compatibility e90acaa602fd

PV yield

Annual yield, range over 2005–2023	9,774 – 11,802 kWh
Mean annual yield	10,663 kWh
Specific yield	977 – 1,180 kWh/kWp
Source	PVGIS 5 (PVGIS-SARAH3) through pvlib, system loss 14 %

Storage sizing

PV at least 0.5 kWp per MWh of annual consumption (HTW Berlin)	2 kWp per MWh, at least 0.5 · met
Usable capacity at most 1.5 kWh per kWp (HTW Berlin)	1.5 kWh × 10 kWp = 15 kWh
Usable capacity at most 1.5 kWh per MWh of annual consumption (HTW Berlin)	1.5 kWh × 5 MWh = 7.5 kWh
Usable capacity, HTW upper limit	7.5 kWh

Self-sufficiency and self-consumption

No battery is selected yet, so the figures assume the storage sizing's upper limit of 7.5 kWh.

Self-sufficiency with storage	67 % – 77 %
Self-consumption with storage	36 % – 41 %
Self-sufficiency without storage	31 % – 41 %
Self-consumption without storage	15 % – 20 %
Storage capacity in these figures	7.5 kWh
Method	HTW Berlin Unabhängigkeitsrechner tables; the range covers the spread of real households (most within 5 percentage points of the table value)

Bill of materials

No bill of materials yet: no released system matches every rule for these inputs.

Yield: PVGIS © European Union · Self-sufficiency: HTW Berlin · Postcodes: GeoNames (CC BY 4.0)

Assumptions and limits

This report contains no economics: no payback period, cash flow or feed-in tariff.

Self-sufficiency figures describe an average household; measured values differ with each household's habits.

Without a roof plane, the yield assumes 35° tilt facing south (HTW Berlin calculator default).

Open-circuit voltage checked at −10 °C cell temperature. (Planning assumption without a published source)

MPP voltage checked at 70 °C cell temperature, with the module's Voc temperature coefficient. (Planning assumption without a published source)

DC cable: 15 m per string and pole until the route is measured. (Planning assumption without a published source)

Two connector pairs per string: one at the array, one at the inverter. (Planning assumption without a published source)

Next steps

Add the Linker products to your solution draft and request a Proforma; we reply within one business day.

Before ordering, confirm the string layout with the module you install, the cable routes and the grid connection on site.

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